# Module 12 — Energy Resolution & Systematics

Supplementary to the sPHENIX Standalone Course

## 1. The stochastic/constant parametrisation

For sampling calorimeters:

```
σ(E)/E  =  a/√E  ⊕  b  ⊕  c/E

  a  stochastic term
  b  constant term
  c  noise term (≈0 in MC)
⊕ means quadrature sum
```

At high energy the constant term dominates (pileup, calibration, leakage). At low energy the stochastic term dominates (sampling + photo-statistics).

## 2. The energy scan

| eV) | Events | Rough run time |
|---|---|---|
| | 10,000 | 2 min |
| | 10,000 | 3 min |
| | 10,000 | 6 min |
| | 10,000 | 10 min |
| | 10,000 | 18 min |
| | 10,000 | 40 min |
| | 5,000 | 35 min |

## 3. Fit workflow

```
// Step 1: for each energy, fit a Gaussian
for (auto& [E, h] : hists) {
    h->Fit("gaus", "Q");
    auto fn = h->GetFunction("gaus");
    double mu = fn->GetParameter(1);
    double sg = fn->GetParameter(2);
    g->SetPoint(n, E, sg/mu);
    g->SetPointError(n, 0, fn->GetParError(2)/mu);
    ++n;
}
// Step 2: fit σ/E = √(a²/E + b²)
TF1 res("res", "sqrt(([0]*[0])/x + [1]*[1])", 0.5, 110);
res.SetParameters(0.10, 0.02);
g->Fit(&res);
```

## 4. Systematic variations

| | How to vary | Expected $\Delta a$, $\Delta b$ |
|---|---|---|
| action | ±10% absorber thickness | $\Delta a/a \approx$ ■5%, $\Delta b$ small |
| cuts | 0.1 mm vs 1 mm | $\Delta a \leq$ few%, $\Delta b \leq 0.1\%$ |
| | FTFP_BERT vs QGSP_BERT | $\Delta a/a \leq 1\%$ for EM |
| | 0 vs 0.126 mm/MeV | Non-linearity at low E |
| ion | Centre vs edge | $\Delta b$ up to 2% (edge leakage) |
| ad | $\sigma_x$ = 0 vs 3 mm | Small for EMCal |

## 5. Reporting

A complete resolution measurement has:

- One $\sigma/E$ vs E plot with the fit overlaid (log-x axis is conventional).
- A table of a, b for each systematic variation.
- Quadrature-summed systematic band added to the nominal quote.
- Statistical uncertainties on each bin ($\sigma$ error bars).
- Comments on residuals (if the fit is bad, quote a high-energy constant value instead).

```
Nominal:            a = (8.4 ± 0.2)%  b = (1.9 ± 0.3)%
+10% absorber:      a = (8.7 ± 0.2)%  b = (1.9 ± 0.3)%
-10% absorber:      a = (8.0 ± 0.2)%  b = (2.0 ± 0.3)%
1 mm cuts:          a = (8.5 ± 0.2)%  b = (1.9 ± 0.3)%
QGSP_BERT:          a = (8.4 ± 0.2)%  b = (1.9 ± 0.3)%
-----------------------------------------------
Systematic quad-sum: Δa = ±0.4    Δb = ±0.1
Final:              a = (8.4 ± 0.2(stat) ± 0.4(syst))%
                    b = (1.9 ± 0.3(stat) ± 0.1(syst))%
```

## 6. Statistical floor

The statistical uncertainty on $\sigma$ from a Gaussian fit with N events is $\sigma(\sigma)/\sigma \approx 1/\sqrt{(2N)}$. For 10,000 events, $\sigma(\sigma)/\sigma \approx 0.7\%$. Any systematic shift smaller than this is noise, not signal.

## 7. Diagnostic plots to produce

| Plot | What it reveals |
|---|---|
| Reconstructed E / True E vs True E | Non-linearity (Birks', leakage, saturation) |
| $\sigma$/E vs E | Main result |
| Fit pull distribution per energy | Is the Gaussian fit adequate? |
| Residual = (E_fit – $\sigma$/E_measured)/σ_err | Goodness of the stochastic-constant fit |

## 8. Beyond electrons: hadrons

Hadronic resolution is typically 50–80%/$\sqrt{E}$ with b ≈ 5–10%. The FTFP_BERT list handles charged pions well. If you want a hadronic study, replace e- with pi- in /gun/particle and expect:

| | Typical a | Typical b |
|---|---|---|
| MCal) | 8–15% | 1–2% |
| Cal) | 8–15% | 1–2% |
| on (HCal) | 50–80% | 5–10% |
| al) | Similar to pion, some offset | 5–10% |

## 9. What a good analysis report includes

• Definition of 'measured energy' (active sum, calibrated, with/without leakage correction).

• Plot of response linearity (intercept and slope) before resolution.

• Fit range and reason for excluding any points.

• Sensitivity study of the fit range.

• Source of each systematic estimate and sample size used.

## 10. Numerical cross-check

For Pb/lAr calorimeter with $f \approx 0.14$, $a \approx 10\%/\sqrt{E}$ is consistent with ATLAS-style LAr performance. The B4a example reproduces this within 10% with default settings.